



TPDN90G18 V. 130A RELEASE NOTE

- ☒ Adopt new version for all new tape out and all designs immediately
 ☐ Adopt new version for all designs, both on going and new start
 ☐ Adopt new version for new design start
 ☐ No action required

1. LIBRARY INFORMATION

Library Name: *TPDN90G18*

Library Version: 130

Release Note Version: A

Library Type: Standard I/O

Technology: *90NM LOGIC GENERAL PURPOSE 1P9M SALICIDE 1.0V/1.8V (CLN90G)*

Library Status: *Silicon Proven*

Design Rule: T-N90-LO-DR-001 v1.3

Spice model: T-N90-LO-SP-001 v1.3

DRC runset: T-N90-LO-DR-001-C1 v1.3a

LVS runset: T-N90-CL-LS-001-C1 v1.2b

LPE runset: T-N90-LO-SP-002-B1 v1.4b

Cell number: 31 (18 I/O cells + 1 corner +6 fillers + 6 bonding pads)

2. UPDATE HISTORY

Date	Version	Changes	Customers action recommended
12/22/04	130A	1. Update the library to comply with design rule V1.3	Adopt new version for all new tape out and all designs immediately
2/24/04	110A	1. First release	Adopt new version for all new tape out and all designs immediately
		2. Silicon proven.	

3. DESIGN KIT VERSION DEPENDENCY TABLE

The following table shows the valid combinations of design-kit versions

<i>rln</i>	<i>gds</i>	<i>gdf</i>	<i>spi</i>	<i>apf</i>	<i>apt</i>	<i>sef</i>	<i>set</i>	<i>syn</i>	<i>vlg</i>	<i>vit</i>	<i>sdc</i>	<i>mdt</i>	<i>doc</i>	<i>lpe</i>	<i>ibs</i>	<i>ctc</i>
130a	130a	-	130a	130a	130a	130a	-	130a	130a	130a	-	130a	130a	130a	130a	130a
110a	110a	-	110a	110a	110a	110a	-	110a	110a	110a	-	110a	110a	110a	-	110a

rln = release note

gds = GDSII layout view

gdf = GDSII phantom view

spi = Spice/lvs netlist

apf = Apollo FRAM (phantom) view

apt = Apollo FRAM and CEL (layout) view

sef = Silicon Ensemble LEF (phantom) view

set = Silicon Ensemble LEF and Virtuso (layout) view

syn = Synopsys

vlg = Verilog

vit = Vital (VHDL)

sdc = Star-DC

mdt = Mentor DFT (Fastscan, DFT advisor)

doc = Documents (Datasheets)

lpe = Layout Parasitic Extracted Spice Netlist

ibs = ibis model file

ctc = Celtic CDB view

4. CHARACTERIZATION CONDITIONS

TYPICAL	BEST	WORST	LOW-TEMP
VDDIO=1.8Volt VDDCORE=1.0Volt Temp=25C Process= NMOS: Typical PMOS: Typical	VDDIO=1.98Volt VDDCORE=1.1Volt Temp=0C Process= NMOS: Fast PMOS: Fast	VDDIO=1.62Volt VDDCORE=0.9Volt Temp=125C Process= NMOS: Slow PMOS: Slow	VDDIO=1.98Volt VDDCORE=1.1Volt Temp= -40C Process= NMOS: Fast PMOS: Fast

5. KNOWN PROBLEMS AND LIMITATIONS

- a. Please do not use smaller filler cells to fill large I/O spacing. DRC errors will be reported if small fillers are inserted to large spacing.
- b. TLF files were derived from Synopsys .lib files; however, the CT-GEN requires a non-decreasing trend in delay table when loading increases. Therefore, the TLF files in this library had been fixed to meet this requirement. Users can generate his/her own TLF files directly from the Synopsys .lib files using the Cadence utility "syn2tlf" if needed.
- c. All types of resistors are recognized as user defined devices (X-card) in new layout versus schematic check (LVS, V1.0). Please include the file 'source_added' provided by LVS runset into the customers' final source netlist when performing LVS check.

6. SPECIAL NOTES

- a. Please use V04.40-PO31 or newer version of celtic for cdb kits.
- b. The library contains a cell named PVDD2POC_18 for POC18 (power on control) signal generation.
It is MANDATORY to implement “one and only one” PVDD2POC_18 cell per separated digital I/O domain. This can be achieved by replacing “one” PVDD2CDG_18 cell with “one” PVDD2POC_18 cell per each separated digital I/O domain. Forgetting to do so would result in a floating POC18 rail, which would affect the functions of I/O buffers, and would in turn jeopardize the whole chip performance.
- c. Spice netlist tpdn90g18_1_2.spi is for separated GND (VSS and VSSPST18) application, and tpdn90g18_3.spi is suitable for same core and I/O GND application.

7. LIBRARY CONTENTS

Cell List: (18 I/O cells + 1 corner + 6 fillers + 6 bonding pads)

I/O Cells: pdc1224cdg_18 pdc0816cdg_18 pdc0204cdg_18 pds1224cdg_18 pds0816cdg_18 pds0204cdg_18 prc1224cdg_18 prc0816cdg_18 prs1224cdg_18 prs0816cdg_18 pvdd1cdg_18 pvdd2cdg_18 pvdd2poc_18 pvss1cdg_18 pvss2cdg_18 pvss3cdg_18 pxoe1cdg_18 pxoe2cdg_18	Corner Cell: pcornerg_18	Pad Cells: pad40n_18 pad40g_18 pad45n_18 pad45g_18 pad60g_18 pad85g_18	Fillers: pfeed0005g_18 pfeed05g_18 pfeed1g_18 pfeed5g_18 pfeed10g_18 pfeed20g_18
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8. TAPE-OUT LAYER INFORMATION

- Please closely follow TSMC 90nm CMOS Logic 1P9M Salicide 1.0V/1.8V Masking Layers and Bias document.
- Please refer to Chapter of “Library Integration Notes” of TSMC I/O Application Note for general tape-out information.
- It is required to tape out with layers listed in the following table. For the rest of layers necessary for tape out, please refer to TSMC 90nm CMOS Logic 1P9M Salicide 1.0V/1.8V Masking Layers and Bias document.

(Continued on the following page)

TSMC STANDARD I/O LIBRARY TPDN90G18

Tape Out Layer Name	Layer Number			
	6ML Tape Out	7ML Tape Out	8ML Tape Out	9ML Tape Out
NWELL	3	3	3	3
OD	6	6	6	6
NT_N	11	11	11	11
OD_18	16	16	16	16
POLYG	17	17	17	17
PIMP	25	25	25	25
NIMP	26	26	26	26
RPO	29	29	29	29
CONT	30	30	30	30
Metal 1	31	31	31	31
Via 1	51	51	51	51
Metal 2	32	32	32	32
Via 2	52	52	52	52
Metal 3	33	33	33	33
Via 3	53	53	53	53
Metal 4	34	34	34	34
Via 4	54	54	54	54
Metal 5	35	35	35	35
Via 5	55	55	55	55
Metal 6	36	36	36	36
Via 6	-	56	56	56
Metal 7	-	37	37	37
Via 7	-	-	57	57
Metal 8	-	-	38	38
Via 8	-	-	-	58
Metal 9	-	-	-	39
Passivation (CB)	43	43	43	43
RHDMY	117	117	117	117
ESD3DMY	147	147	147	147